CSIE 2344, Spring 2026: Laboratory 1

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1 Source Code

1.1

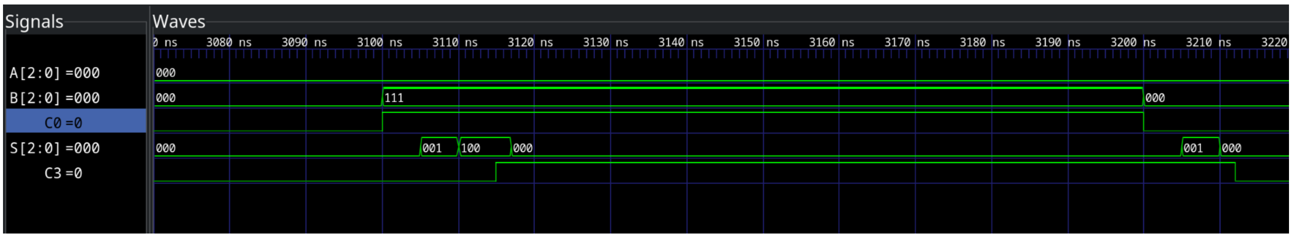
```verilog
module rca_gl(
        output C3,        // carry output
        output[2:0] S,    // sum
        input[2:0] A, B, // operands
        input C0          // carry input
        );

        wire c1, c2;
        FA fa0(c1, S[0], A[0], B[0], C0);
        FA fa1(c2, S[1], A[1], B[1], c1);
        FA fa2(C3, S[2], A[2], B[2], c2);

endmodule
```

1.2

```verilog
module cla_gl(
    output C3,        // carry output
    output[2:0] S,    // sum
    input[2:0] A, B, // operands
    input C0          // carry input
    );

    wire [2:0] g, p, p_xor;
    wire c1, c2;

    AND g_and0(g[0], A[0], B[0]);
    AND g_and1(g[1], A[1], B[1]);
```

```
    AND g_and2(g[2], A[2], B[2]);

    OR p_or0(p[0], A[0], B[0]);
    OR p_or1(p[1], A[1], B[1]);
    OR p_or2(p[2], A[2], B[2]);

    XOR p_xor0(p_xor[0], A[0], B[0]);
    XOR p_xor1(p_xor[1], A[1], B[1]);
    XOR p_xor2(p_xor[2], A[2], B[2]);

    wire p0c0;
    AND and_p0c0(p0c0, p[0], C0);
    OR or_c1(c1, g[0], p0c0);

    wire g0p1, c0p0p1, c2_tmp;
    AND and_c0p0p1(c0p0p1, p[1], p0c0);
    AND and_g0p1(g0p1, g[0], p[1]);
    OR or_c2tmp(c2_tmp, g[1], g0p1);
    OR or_c2(c2, c2_tmp, c0p0p1);

    wire g1p2, g0p1p2, c0p0p1p2;
    AND4 and4_c0p0p1p2(c0p0p1p2, C0, p[0], p[1], p[2]);
    AND and_g0p1p2(g0p1p2, g0p1, p[2]);
    AND and_g1p2(g1p2, g[1], p[2]);

    OR4 or4_c3(C3, g[2], g1p2, g0p1p2, c0p0p1p2);

    XOR xor_s0(S[0], p_xor[0], C0);
    XOR xor_s1(S[1], p_xor[1], c1);
    XOR xor_s2(S[2], p_xor[2], c2);
endmodule
```

2 Waveform



3 Propagation Delays

3.1

The maximum delay is 23 ticks on transition 000+000+0 → 000+111+1.

3.2

The maximum delay is 17 ticks on transition 000+000+1 → 000+000+0.